

CS 305 Lab Tutorial

Lab11 IP fragment & ICMP

Dept. Computer Science and Engineering
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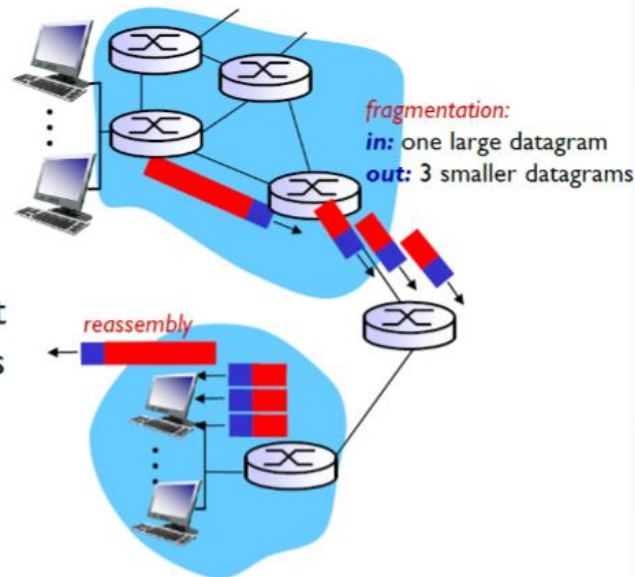
Topic

- IP Fragment
 - IPv4 vs IPv6
- ICMP
 - ICMPv4
 - ICMPv4 Header
 - Examples:
 - echo request, echo reply, Destination unreachable, Time exceeded
 - ICMPv6
 - ICMPv4 Header
 - Examples:
 - echo request, echo reply, Time exceeded, Packet too Big

IPv4 Fragmentation and Reassembly

IP fragmentation, reassembly

- network links have MTU (max.transfer size) - largest possible link-level frame
 - different link types, different MTUs
- large IP datagram divided (“fragmented”) within net
 - one datagram becomes several datagrams
 - “reassembled” only at final destination
 - IP header bits used to identify, order related fragments



IPv4 Fragment related fields in header(1)

Flags: 3 bits

Various Control Flags.

Bit 0: reserved, must be zero

Bit 1: (DF) 0 = May Fragment, 1 = Don't Fragment.

Bit 2: (MF) 0 = Last Fragment, 1 = More Fragments.

	0	1	2
0	0	F	F
1		D	
2			M

- ```

▼ Internet Protocol Version 4, Src: 192.168.2.104 (192.168.2.104), Dst: 116.7.234.3 (116.7.234.3)
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 ▼ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 0000 00.. = Differentiated Services Codepoint: Default (0)
 00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
 Total Length: 1020
 Identification: 0x0a9c (2716)
 ▼ Flags: 0x00b9
 0... = Reserved bit: Not set
 .0.. = Don't fragment: Not set
 ..0. = More fragments: Not set
 ..0 0000 1011 1001 = Fragment offset: 185

```

## Fragment Offset: 13 bits

This field indicates where in the datagram this fragment belongs.

The fragment offset is measured **in units of 8 octets** (64 bits). The first fragment has offset zero.

*Tips in Wireshark: ip.flags.mf*

|                     |   |   |   |     |   |   |   |                 |   |   |   |                 |   |   |   |         |   |   |   |                 |   |  |  |  |  |  |  |
|---------------------|---|---|---|-----|---|---|---|-----------------|---|---|---|-----------------|---|---|---|---------|---|---|---|-----------------|---|--|--|--|--|--|--|
| 0                   |   |   |   | 1   |   |   |   | 2               |   |   |   | 3               |   |   |   |         |   |   |   |                 |   |  |  |  |  |  |  |
| 0                   | 1 | 2 | 3 | 4   | 5 | 6 | 7 | 8               | 9 | 0 | 1 | 2               | 3 | 4 | 5 | 6       | 7 | 8 | 9 | 0               | 1 |  |  |  |  |  |  |
| Version             |   |   |   | IHL |   |   |   | Type of Service |   |   |   | Total Length    |   |   |   |         |   |   |   |                 |   |  |  |  |  |  |  |
| Identification      |   |   |   |     |   |   |   |                 |   |   |   |                 |   |   |   | Flags   |   |   |   | Fragment Offset |   |  |  |  |  |  |  |
| Time to Live        |   |   |   |     |   |   |   | Protocol        |   |   |   | Header Checksum |   |   |   |         |   |   |   |                 |   |  |  |  |  |  |  |
| Source Address      |   |   |   |     |   |   |   |                 |   |   |   |                 |   |   |   |         |   |   |   |                 |   |  |  |  |  |  |  |
| Destination Address |   |   |   |     |   |   |   |                 |   |   |   |                 |   |   |   |         |   |   |   |                 |   |  |  |  |  |  |  |
| Options             |   |   |   |     |   |   |   |                 |   |   |   |                 |   |   |   | Padding |   |   |   |                 |   |  |  |  |  |  |  |

### Example Internet Datagram Header

# IPv4 Fragment related fields in header(2)

**Identification:** An internet header field carrying the identifying value assigned by the sender to aid in assembling the fragments of a datagram.

| No.                                                                                                         | Time     | Source        | Destination | Protocol | Le Info                                                    |
|-------------------------------------------------------------------------------------------------------------|----------|---------------|-------------|----------|------------------------------------------------------------|
| 1                                                                                                           | 0.000000 | 192.168.2.104 | 47.75.42.25 | IPv4     | 1... Fragmented IP protocol (proto=ICMP 1, off=0, ID=e6be) |
| <                                                                                                           |          |               |             |          |                                                            |
| > Frame 1: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \Device\NPF_{...} |          |               |             |          |                                                            |
| > Ethernet II, Src: IntelCor_5..., Dst: Skyworth_de:ad:05 (00:1a:9a:de:ad:05)                               |          |               |             |          |                                                            |
| ▼ Internet Protocol Version 4, Src: 192.168.2.104, Dst: 47.75.42.25                                         |          |               |             |          |                                                            |
| 0100 .... = Version: 4                                                                                      |          |               |             |          |                                                            |
| .... 0101 = Header Length: 20 bytes (5)                                                                     |          |               |             |          |                                                            |
| > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)                                             |          |               |             |          |                                                            |
| Total Length: 1500                                                                                          |          |               |             |          |                                                            |
| Identification: 0xe6be (59070)                                                                              |          |               |             |          |                                                            |
| ▼ Flags: 0x2000, More fragments                                                                             |          |               |             |          |                                                            |
| 0... .. = Reserved bit: Not set                                                                             |          |               |             |          |                                                            |
| .0.. .. = Don't fragment: Not set                                                                           |          |               |             |          |                                                            |
| ..1. .... = More fragments: Set                                                                             |          |               |             |          |                                                            |
| Fragment offset: 0                                                                                          |          |               |             |          |                                                            |
| Time to live: 64                                                                                            |          |               |             |          |                                                            |
| Protocol: ICMP (1)                                                                                          |          |               |             |          |                                                            |
| Header checksum: 0x51ee [validation disabled]                                                               |          |               |             |          |                                                            |
| [Header checksum status: Unverified]                                                                        |          |               |             |          |                                                            |
| Source: 192.168.2.104                                                                                       |          |               |             |          |                                                            |
| Destination: 47.75.42.25                                                                                    |          |               |             |          |                                                            |
| [Reassembled IPv4 in frame: 2]                                                                              |          |               |             |          |                                                            |
| > Data (1480 bytes)                                                                                         |          |               |             |          |                                                            |

| No.                                                                                                 | Time     | Source        | Destination | Protocol | Le Info                                                           |
|-----------------------------------------------------------------------------------------------------|----------|---------------|-------------|----------|-------------------------------------------------------------------|
| 2                                                                                                   | 0.000000 | 192.168.2.104 | 47.75.42.25 | ICMP     | 62 Echo (ping) request id=0x0001, seq=29/7424, ttl=64 (reply in 4 |
| <                                                                                                   |          |               |             |          |                                                                   |
| > Frame 2: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface \Device\NPF_{...} |          |               |             |          |                                                                   |
| > Ethernet II, Src: IntelCor_5..., Dst: Skyworth_de:ad:05 (00:1a:9a:de:ad:05)                       |          |               |             |          |                                                                   |
| ▼ Internet Protocol Version 4, Src: 192.168.2.104, Dst: 47.75.42.25                                 |          |               |             |          |                                                                   |
| 0100 .... = Version: 4                                                                              |          |               |             |          |                                                                   |
| .... 0101 = Header Length: 20 bytes (5)                                                             |          |               |             |          |                                                                   |
| > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)                                     |          |               |             |          |                                                                   |
| Total Length: 48                                                                                    |          |               |             |          |                                                                   |
| Identification: 0xe6be (59070)                                                                      |          |               |             |          |                                                                   |
| ▼ Flags: 0x00b9                                                                                     |          |               |             |          |                                                                   |
| 0... .. = Reserved bit: Not set                                                                     |          |               |             |          |                                                                   |
| .0.. .. = Don't fragment: Not set                                                                   |          |               |             |          |                                                                   |
| ..0. .... = More fragments: Not set                                                                 |          |               |             |          |                                                                   |
| Fragment offset: 1480                                                                               |          |               |             |          |                                                                   |
| Time to live: 64                                                                                    |          |               |             |          |                                                                   |
| Protocol: ICMP (1)                                                                                  |          |               |             |          |                                                                   |
| Header checksum: 0x76e1 [validation disabled]                                                       |          |               |             |          |                                                                   |
| [Header checksum status: Unverified]                                                                |          |               |             |          |                                                                   |
| Source: 192.168.2.104                                                                               |          |               |             |          |                                                                   |
| Destination: 47.75.42.25                                                                            |          |               |             |          |                                                                   |
| > [2 IPv4 Fragments (1508 bytes): #1(1480), #2(28)]                                                 |          |               |             |          |                                                                   |
| > Internet Control Message Protocol                                                                 |          |               |             |          |                                                                   |

Tips in Wireshark : ip.id

|                           |   |   |   |                     |   |   |   |                 |   |   |   |              |   |   |   |                 |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |
|---------------------------|---|---|---|---------------------|---|---|---|-----------------|---|---|---|--------------|---|---|---|-----------------|---|---|---|-----------------|---|--|--|--|--|--|--|--|--|--|--|
| 0                         |   |   |   | 1                   |   |   |   | 2               |   |   |   | 3            |   |   |   |                 |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |
| 0                         | 1 | 2 | 3 | 4                   | 5 | 6 | 7 | 8               | 9 | 0 | 1 | 2            | 3 | 4 | 5 | 6               | 7 | 8 | 9 | 0               | 1 |  |  |  |  |  |  |  |  |  |  |
| +-----+-----+-----+-----+ |   |   |   |                     |   |   |   |                 |   |   |   |              |   |   |   |                 |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |
| Version                   |   |   |   | IHL                 |   |   |   | Type of Service |   |   |   | Total Length |   |   |   |                 |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |
| +-----+-----+-----+-----+ |   |   |   |                     |   |   |   |                 |   |   |   |              |   |   |   |                 |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |
| Identification            |   |   |   |                     |   |   |   |                 |   |   |   | Flags        |   |   |   | Fragment Offset |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |
| +-----+-----+-----+-----+ |   |   |   |                     |   |   |   |                 |   |   |   |              |   |   |   |                 |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |
|                           |   |   |   | Time to Live        |   |   |   |                 |   |   |   | Protocol     |   |   |   |                 |   |   |   | Header Checksum |   |  |  |  |  |  |  |  |  |  |  |
| +-----+-----+-----+-----+ |   |   |   |                     |   |   |   |                 |   |   |   |              |   |   |   |                 |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |
|                           |   |   |   | Source Address      |   |   |   |                 |   |   |   |              |   |   |   |                 |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |
| +-----+-----+-----+-----+ |   |   |   |                     |   |   |   |                 |   |   |   |              |   |   |   |                 |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |
|                           |   |   |   | Destination Address |   |   |   |                 |   |   |   |              |   |   |   |                 |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |
| +-----+-----+-----+-----+ |   |   |   |                     |   |   |   |                 |   |   |   |              |   |   |   |                 |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |
|                           |   |   |   | Options             |   |   |   |                 |   |   |   |              |   |   |   |                 |   |   |   | Padding         |   |  |  |  |  |  |  |  |  |  |  |
| +-----+-----+-----+-----+ |   |   |   |                     |   |   |   |                 |   |   |   |              |   |   |   |                 |   |   |   |                 |   |  |  |  |  |  |  |  |  |  |  |

Example Internet Datagram Header

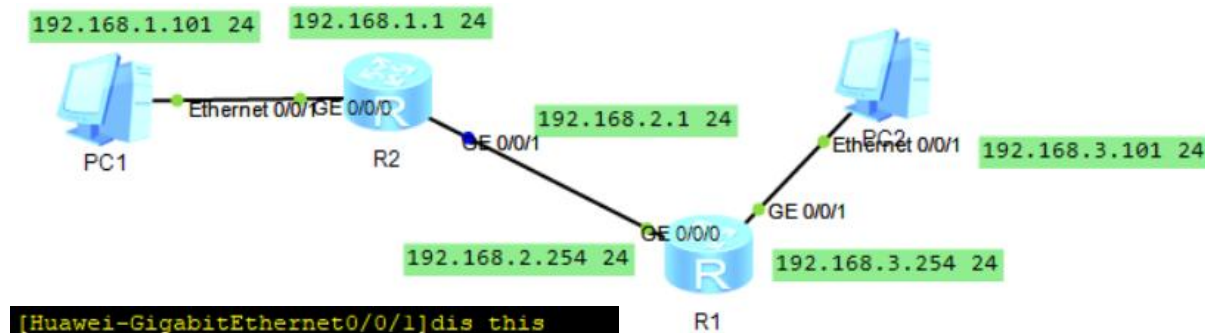
# Demo1 - IPv4 fragment(1)

Build the network as the topology shown on the right hand, after the configuration, PC1 could send/receive the pkt to/from PC2.

**Tips: in eNSP, router support IP fragment while PC don't.**

Set the the MTU of R2's interface GE0/0/1 as 50.

Invoke "ping" test on R2, and capture the packet ont its interface GE 0/0/1



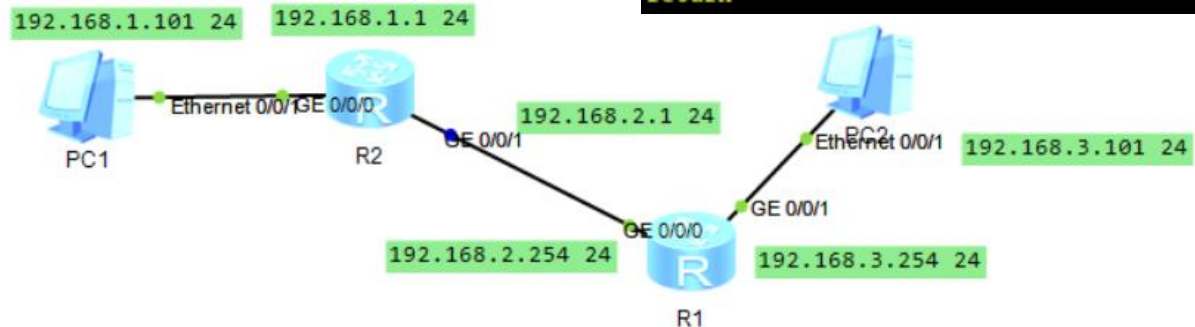
```
[Huawei-GigabitEthernet0/0/1]dis this
#
interface GigabitEthernet0/0/1
mtu 50
ip address 192.168.2.1 255.255.255.0
#
return
```

```
R2
[Huawei]ping 192.168.3.254
PING 192.168.3.254: 56 data bytes, press CTRL_C to break
Reply from 192.168.3.254: bytes=56 Sequence=1 ttl=255 time=60 ms
Reply from 192.168.3.254: bytes=56 Sequence=2 ttl=255 time=30 ms
Reply from 192.168.3.254: bytes=56 Sequence=3 ttl=255 time=10 ms
Reply from 192.168.3.254: bytes=56 Sequence=4 ttl=255 time=50 ms
Reply from 192.168.3.254: bytes=56 Sequence=5 ttl=255 time=40 ms
```

Tips: in eNSP, route support IP fragment while PC don't.

```
[Huawei-GigabitEthernet0/0/1]dis this
#
interface GigabitEthernet0/0/1
mtu 50
ip address 192.168.2.1 255.255.255.0
#
return
```

Invoke “ping” test on R2, and capture the packet on its interface GE 0/0/1



| Time       | Source        | Protocol | Destination   | Length | Info                                                                       |
|------------|---------------|----------|---------------|--------|----------------------------------------------------------------------------|
| 1 0.000000 | 192.168.2.1   | IPv4     | 192.168.3.254 | 60     | Fragmented IP protocol (proto=ICMP 1, off=0, ID=001c) [Reassembled in #3]  |
| 2 0.000000 | 192.168.2.1   | IPv4     | 192.168.3.254 | 60     | Fragmented IP protocol (proto=ICMP 1, off=24, ID=001c) [Reassembled in #3] |
| 3 0.000000 | 192.168.2.1   | ICMP     | 192.168.3.254 | 60     | Echo (ping) request id=0xd0ab, seq=256/1, ttl=255 (reply in 4)             |
| 4 0.015000 | 192.168.3.254 | ICMP     | 192.168.2.1   | 98     | Echo (ping) reply id=0xd0ab, seq=256/1, ttl=255 (request in 3)             |

```
[3 IPv4 Fragments (64 bytes): #1(24), #2(24), #3(16)]
[Frame: 1, payload: 0-23 (24 bytes)]
[Frame: 2, payload: 24-47 (24 bytes)]
[Frame: 3, payload: 48-63 (16 bytes)]
[Fragment count: 3]
[Reassembled IPv4 length: 64]
[Reassembled IPv4 data: 080075d9d0ab01007e380800000000]
```

Q1: As shown in the packets 1-4, only the packet send by 192.168.2.1 is fraged, the packet it received is not, why?

Q2: How does the Wireshak identify the order of each fragment in an IP packet ?



# IPv6 Fragment (1)

Unlike IPv4, fragmentation in IPv6 is performed only by source nodes, NOT by routers along a packet's delivery path -- see [RFC8200]

- **Fragment header:** The Fragment header is used by an IPv6 source to send a packet larger than would fit in the path MTU to its destination.
- The Fragment header is identified by a Next Header value of **44** in the immediately preceding header and has the following format.

Tips:

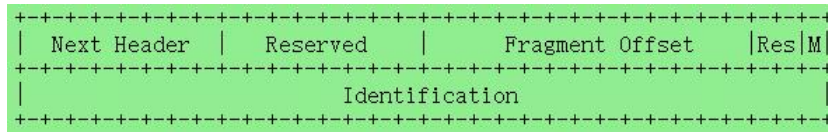
using 'ping -6 www.sustech.edu.cn -l 2000' to invoke an ICMPv6 transaction, there would be a **Fragment header** as an **extension Header** in the IPv6 packet.

|                |          |                 |         |
|----------------|----------|-----------------|---------|
| Next Header    | Reserved | Fragment Offset | [Res M] |
| Identification |          |                 |         |

| No.                                                                                                      | Time     | Source         | Protocol | Destination   | Info                                                 |
|----------------------------------------------------------------------------------------------------------|----------|----------------|----------|---------------|------------------------------------------------------|
| 3                                                                                                        | 2.911236 | 2001:da8:20... | IPv6     | 2001:da8:2... | IPv6 fragment (off=0 more=y ident=0x558ba252 nxt=58) |
| <                                                                                                        |          |                |          |               |                                                      |
| > Frame 3: 1486 bytes on wire (11888 bits), 1486 bytes captured (11888 bits) on interface \Device\NPF... |          |                |          |               |                                                      |
| > Ethernet II, Src: IntelCor_d2:20:30 (04:ed:33:d2:20:30), Dst: JuniperN_d0:93:c1 (3c:8c:93:d0:93:c1)    |          |                |          |               |                                                      |
| v Internet Protocol Version 6, Src: 2001:da8:201d:1109::fcd7, Dst: 2001:da8:201d::42:3                   |          |                |          |               |                                                      |
| 0110 .... = Version: 6                                                                                   |          |                |          |               |                                                      |
| > .... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)                                    |          |                |          |               |                                                      |
| .... 0000 0000 0000 0000 0000 = Flow Label: 0x000000                                                     |          |                |          |               |                                                      |
| Payload Length: 1432                                                                                     |          |                |          |               |                                                      |
| Next Header: Fragment Header for IPv6 (44)                                                               |          |                |          |               |                                                      |
| Hop Limit: 128                                                                                           |          |                |          |               |                                                      |
| Source Address: 2001:da8:201d:1109::fcd7                                                                 |          |                |          |               |                                                      |
| Destination Address: 2001:da8:201d::42:3                                                                 |          |                |          |               |                                                      |
| v Fragment Header for IPv6                                                                               |          |                |          |               |                                                      |
| Next header: ICMPv6 (58)                                                                                 |          |                |          |               |                                                      |
| Reserved octet: 0x00                                                                                     |          |                |          |               |                                                      |
| 0000 0000 0000 0... = Offset: 0 (0 bytes)                                                                |          |                |          |               |                                                      |
| .... .00. = Reserved bits: 0                                                                             |          |                |          |               |                                                      |
| .... .1 = More Fragments: Yes                                                                            |          |                |          |               |                                                      |
| Identification: 0x558ba252                                                                               |          |                |          |               |                                                      |
| <a href="#">[Reassembled IPv6 in frame: 4]</a>                                                           |          |                |          |               |                                                      |
| > Data (1424 bytes)                                                                                      |          |                |          |               |                                                      |

# IPv6 Fragment (2)

'ping -6 www.sustech.edu.cn -l 2000'



- IPv6.data(2008) = ICMPv6.header(8) + ICMPv6.data(2000)
- IPv6.data(2008) = data1(1424) + data2(584)
- IPv6.fragment1.data(1432) = extensionHeader(8)+1424(data1)
- IPv6.fragment2.data(592) = extensionHeader(8)+584(data2)

```

No. Time Source Protocol Destination Info
3 2.911236 2001:da8:20... IPv6 2001:da8:2... IPv6 fragment (off=0 more=y ident=0x558b)
<
> Frame 3: 1486 bytes on wire (11888 bits), 1486 bytes captured (11888 bits) on interface \Device\NPF_{7E...}
> Ethernet II, Src: IntelCor_d2:20:30 (04:ed:33:d2:20:30), Dst: JuniperN_d0:93:c1 (3c:8c:93:d0:93:c1)
> Internet Protocol Version 6, Src: 2001:da8:201d:1109::fcd7, Dst: 2001:da8:201d::42:3
 0110 = Version: 6
 > 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
 0000 0000 0000 0000 = Flow Label: 0x000000
 Payload Length: 1432
 Next Header: Fragment Header for IPv6 (44)
 Hop Limit: 128
 Source Address: 2001:da8:201d:1109::fcd7
 Destination Address: 2001:da8:201d::42:3
 > Fragment Header for IPv6
 Next header: ICMPv6 (58)
 Reserved octet: 0x00
 0000 0000 0000 0... = Offset: 0 (0 bytes)
 0000 0000 .00. = Reserved bits: 0
 0000 0000 ...1 = More Fragments: Yes
 Identification: 0x558ba252
 [Reassembled IPv6 in frame: 4]
 > Data (1424 bytes)

```

```

No. Time Source Protocol Destination Info
4 2.911236 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) request id=0x0001, seq=9, hop limit=128
<
> Frame 4: 646 bytes on wire (5168 bits), 646 bytes captured (5168 bits) on interface \Device\NPF_{7E...}
> Ethernet II, Src: IntelCor_d2:20:30 (04:ed:33:d2:20:30), Dst: JuniperN_d0:93:c1 (3c:8c:93:d0:93:c1)
> Internet Protocol Version 6, Src: 2001:da8:201d:1109::fcd7, Dst: 2001:da8:201d::42:3
 0110 = Version: 6
 > 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
 0000 0000 0000 0000 = Flow Label: 0x000000
 Payload Length: 592
 Next Header: Fragment Header for IPv6 (44)
 Hop Limit: 128
 Source Address: 2001:da8:201d:1109::fcd7
 Destination Address: 2001:da8:201d::42:3
 > Fragment Header for IPv6
 Next header: ICMPv6 (58)
 Reserved octet: 0x00
 0000 0101 1001 0... = Offset: 178 (1424 bytes)
 0000 0000 .00. = Reserved bits: 0
 0000 0000 ...0 = More Fragments: No
 Identification: 0x558ba252
 > [2 IPv6 Fragments (2008 bytes): #3(1424), #4(584)]
 [Frame: 3, payload: 0-1423 (1424 bytes)]
 [Frame: 4, payload: 1424-2007 (584 bytes)]
 [Fragment count: 2]
 [Reassembled IPv6 length: 2008]
 [Reassembled IPv6 data: 800051a9000100096162636465666768696a6b6c6d6e6f707172737475767761626364]
 > Internet Control Message Protocol v6

```

# Part B. ICMP

- ICMP is used from gateways to hosts and between hosts to report errors and make routing suggestions.
- ICMP and IP :
  - Internet protocol errors may be reported via the ICMP messages
  - **ICMP uses the basic support of IP as if it were a higher level protocol, however, ICMP is actually an integral part of IP, and must be implemented by every IP module.**

<https://datatracker.ietf.org/doc/html/rfc792>

# ICMPv4-examples(1)Echo Request, Echo Reply

- The data received in the echo message must be returned in the echo reply message.
- Type
  - **8** for **echo request** message;
  - **0** for **echo reply** message.
- Code
  - 0
- The identifier and sequence number may be used by the echo sender to aid in matching the replies with the echo requests. The echoer returns these same values in the echo reply.

### Echo or Echo Reply Message

[illegible]

# ICMPv4-examples(1): Echo Request

Initial the session with following cmd command:

ping www.sustech.edu.cn

```
ip.proto==1
No. Time Source Destination
8015 68.531009 192.168.2.104 www.sustech.edu.cn.w.cdngslb.com
8016 68.554768 www.sustech.edu.cn.w... 192.168.2.104

Frame 8015: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface
Ethernet II, Src: IntelCor_5c:69:58 (90:61:ae:5c:69:58), Dst: Skyworth_de:ad:05 (00
Internet Protocol Version 4, Src: 192.168.2.104 (192.168.2.104), Dst: www.sustech.e
0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 60
Identification: 0xa295 (41621)
Flags: 0x0000
0... .. = Reserved bit: Not set
.0... .. = Don't fragment: Not set
..0... .. = More fragments: Not set
...0 0000 0000 0000 = Fragment offset: 0
Time to live: 64
Protocol: ICMP (1)
Header checksum: 0xc561 [validation disabled]
[Header checksum status: Unverified]
Source: 192.168.2.104 (192.168.2.104)
Destination: www.sustech.edu.cn.w.cdngslb.com (183.232.151.209)
Internet Control Message Protocol
Type: 8 (Echo (ping) request)
Code: 0
Checksum: 0x4c5e [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence number (BE): 253 (0x00fd)
Sequence number (LE): 64768 (0xfd00)
[Response frame: 8016]
Data (32 bytes)
Data: 6162636465666768696a6b6c6d6e6f707172737475767761...
```

ICMP. type : 8

ICMP. code : 0

*Tips in Wireshark :*

*ip.proto == 1 or ICMP.type*

*Q1. What's the size of this ICMP Header, ICMP message and this IP packet?*

*Q2. What's value of Identifier and Sequence number field in this ICMP message?*

Echo or Echo Reply Message

|                                                                 |                 |           |           |
|-----------------------------------------------------------------|-----------------|-----------|-----------|
| 0                                                               | 1               | 2         | 3         |
| 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 |                 |           |           |
| ++-----++                                                       | ++-----++       | ++-----++ | ++-----++ |
| Type                                                            | Code            | Checksum  |           |
| ++-----++                                                       | ++-----++       | ++-----++ | ++-----++ |
| Identifier                                                      | Sequence Number |           |           |
| ++-----++                                                       | ++-----++       | ++-----++ | ++-----++ |
| Data ...                                                        |                 |           |           |
| ++-----++                                                       |                 |           |           |

# ICMPv4-examples(1): Echo Reply

```
ip.proto==1
No. Time Source Destination
8016 68.554768 www.sustech.edu.cn.w... 192.168.2.104

> Ethernet II, Src: Skyworth_de:ad:05 (00:1a:9a:de:ad:05), Dst: IntelCor_5c:69:58 (90:61:ae:5c:69:58)
> Internet Protocol Version 4, Src: www.sustech.edu.cn.w.cdngslb.com (183.232.151.209), Dst: 192.168.2.104
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 60
 Identification: 0xa295 (41621)
 > Flags: 0x0000
 0... .. = Reserved bit: Not set
 .0.. .. = Don't fragment: Not set
 ..0. = More fragments: Not set
 ...0 0000 0000 0000 = Fragment offset: 0
 Time to live: 24
 Protocol: ICMP (1)
 Header checksum: 0xed61 [validation disabled]
 [Header checksum status: Unverified]
 Source: www.sustech.edu.cn.w.cdngslb.com (183.232.151.209)
 Destination: 192.168.2.104 (192.168.2.104)
 > Internet Control Message Protocol
 Type: 0 (Echo (ping) reply)
 Code: 0
 Checksum: 0x545e [correct]
 [Checksum Status: Good]
 Identifier (BE): 1 (0x0001)
 Identifier (LE): 256 (0x0100)
 Sequence number (BE): 253 (0x00fd)
 Sequence number (LE): 64768 (0xfd00)
 [Request frame: 8015]
 [Response time: 23.759 ms]
 > Data (32 bytes)
 Data: 6162636465666768696a6b6c6d6e6f707172737475767761...
 [Length: 32]
```

ICMP. type : 0  
ICMP. code : 0

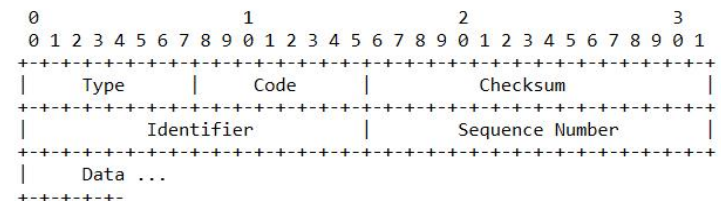
*Tips in Wireshark : ICMP.type*

*Q1. List the differences between the #8015 frame(described on last page) and the #8016 frame(described on this page)*

1. source and destination
2. TTL field of IP Header
3. Total Length field of IP Header
4. ICMP Header

...

Echo or Echo Reply Message



# ICMPv4-examples(2): Destination unreachable(1)

The image shows a Wireshark packet capture of an ICMP message. The top bar indicates the packet is an ICMP message from 192.168.1.1 to 192.168.1.101, with the reason '70 Destination unreachable (Network unreachable)'. The packet details pane shows the following structure:

- Frame 4: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface -, id 0
- Ethernet II, Src: HuaweiTe\_2f:18:51 (54:89:98:2f:18:51), Dst: HuaweiTe\_c8:5b:6d (54:89:98:c8:5b:6d)
- Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.101
- Internet Control Message Protocol
  - Type: 3 (Destination unreachable)
  - Code: 0 (Network unreachable)
  - Checksum: 0x6e81 [correct] [Checksum Status: Good]
  - Unused: 00000000
- Internet Protocol Version 4, Src: 192.168.1.101, Dst: 192.168.3.1
- Internet Control Message Protocol
  - Type: 8 (Echo (ping) request)
  - Code: 0
  - Checksum: 0x0b1e [unverified] [in ICMP error packet] [Checksum Status: Unverified]
  - Identifier (BE): 31583 (0x7b5f)
  - Identifier (LE): 24443 (0x5f7b)
  - Sequence Number (BE): 1 (0x0001)
  - Sequence Number (LE): 256 (0x0100)

Annotations on the right side of the image provide additional context:

- ICMP. type : 3
- ICMP. code : 0
- network unreachable

A note at the bottom right explains the scenario: *A network node (usually a Router) send a ICMP message to the source, the destination is unreachable, the reason could be found in the filed "Code" of ICMP*

```
ICMP. type : 3
ICMP. code : 0
network unreachable
```

*A network node (usually a Router) send a ICMP message to the source, the destination is unreachable, the reason could be found in the filed "Code" of ICMP Header.*

## Destination Unreachable Message

```

0 1 2 3
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
| Type | Code | Checksum |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
| unused |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
| Internet Header + 64 bits of Original Data Datagram |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+

```

### Tips in Wireshark : ICMP.type

## ICMPv4 -examples(2): Destination unreachable(2)

| No. | Time     | Source      | Protocol | Destination   | Length | Info                       |
|-----|----------|-------------|----------|---------------|--------|----------------------------|
| 6   | 1.078000 | 192.168.1.1 | ICMP     | 192.168.1.101 | 70     | Destination unreachable (F |

> Frame 8: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface -, id 0  
 > Ethernet II, Src: HuaweiTe\_2f:18:51 (54:89:98:2f:18:51), Dst: HuaweiTe\_c8:5b:6d (54:89:98:c8:5b:6d)  
 > Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.101  
 > Internet Control Message Protocol  
     Type: 3 (Destination unreachable)  
     Code: 4 (Fragmentation needed)  
     Checksum: 0x6f4b [correct]  
     [Checksum Status: Good]  
     Unused: 0000  
     MTU of next hop: 50  
 > Internet Protocol Version 4, Src: 192.168.1.101, Dst: 192.168.3.1  
 > Internet Control Message Protocol  
     Type: 8 (Echo (ping) request)  
     Code: 0  
     Checksum: 0xb620 [unverified] [in ICMP error packet]  
     [Checksum Status: Unverified]  
     Identifier (BE): 53338 (0xd05a)  
     Identifier (LE): 23248 (0x5ad0)  
     Sequence Number (BE): 3 (0x0003)  
     Sequence Number (LE): 768 (0x0300)

ICMP. type : 3  
 ICMP. code : 4  
 Fragmentation needed

*A network node (usually a Router) send a ICMP message to the source, the destination is unreachable, the reason could be found in the filed "Code" of ICMP*

```
ICMP. type : 3
ICMP. code : 4
Fragmentation needed
```

*A network node (usually a Router) send a ICMP message to the source, the destination is unreachable, the reason could be found in the filed "Code" of ICMP Header.*

## Destination Unreachable Message

```

0 1 2 3
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+
| Type | Code | Checksum |
+-+-+
| unused |
+-+-+
| Internet Header + 64 bits of Original Data Datagram |
+-+-+

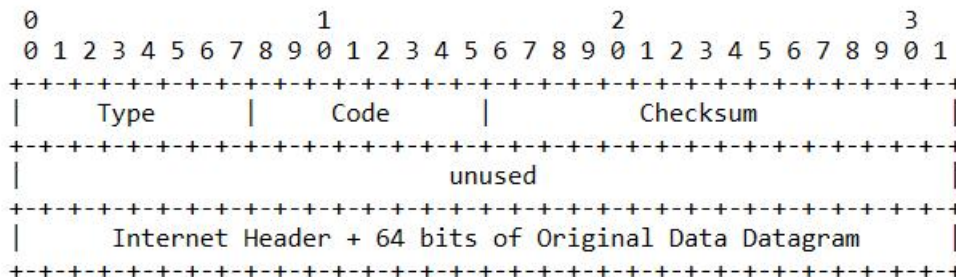
```

### Tips in Wireshark : ICMP.type

# ICMPv4-examples(3):Time Exceeded(1)

Time Exceeded Message

ICMP. type : 11



Code 0 : time to live exceeded in transit;

Code 1 : fragment reassembly time exceeded.

If the gateway processing a datagram finds the time to live field is zero it must discard the datagram. The gateway may also notify the source host via the time exceeded message.

If a host reassembling a fragmented datagram cannot complete the reassembly due to missing fragments within its time limit it discards the datagram, and it may send a time exceeded message.

Code 0 may be received from a gateway. Code 1 may be received from a host.

# ICMPv4-examples(3):Time Exceeded(2)

Initial the session with following cmd: `tracert / traceroute`

```

Internet Protocol Version 4 Src: 192.168.2.1 (192.168.2.1), Dst: 192.168.2.104 (192.168.2.104)
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 0000 00.. = Differentiated Services Codepoint: Default (0)
 00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
 Total Length: 56
 Identification: 0x07cf (1999)
 > Flags: 0x0000
 Time to live: 64
 Protocol: ICMP (1)
 Header checksum: 0xed3c [validation disabled]
 [Header checksum status: Unverified]
 Source: 192.168.2.1 (192.168.2.1)
 Destination: 192.168.2.104 (192.168.2.104)
Internet Control Message Protocol
 Type: 11 (Time-to-live exceeded)
 Code: 0 (Time to live exceeded in transit)
 checksum: 0x101b [correct]
 [Checksum Status: Good]
Internet Protocol Version 4 Src: 192.168.2.104 (192.168.2.104), Dst: 116.7.234.3 (116.7.234.3)
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 1500
 Identification: 0x0a9c (2716)
 > Flags: 0x2000, More fragments
 > Time to live: 1
 Protocol: ICMP (1)
 Header checksum: 0x686a [validation disabled]
 [Header checksum status: Unverified]
 Source: 192.168.2.104 (192.168.2.104)
 Destination: 116.7.234.3 (116.7.234.3)
Internet Control Message Protocol

```

Tips in Wireshark : *ICMP.type*

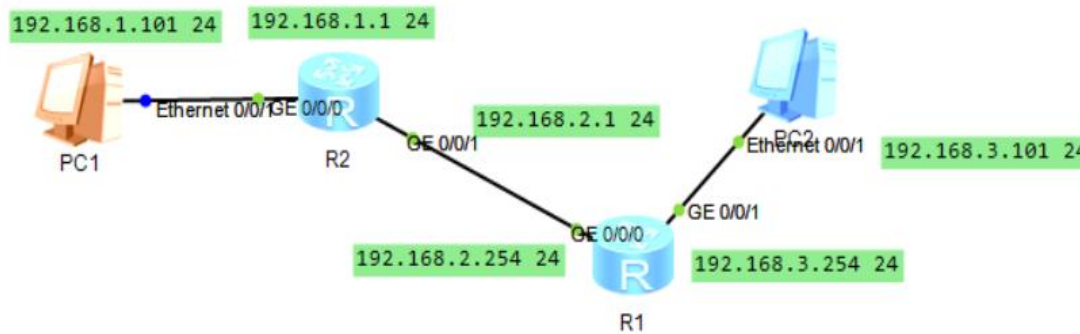
Q1. Is the outside IP's src address same with the inside IP's dest address? Why?

Q2. Is the TTL of outside IP same with which in inside IP? why?

Time Exceeded Message

| 0                                                   |   |   |   |   |   |   |   |   |   | 1    |   |   |   |   |   |   |   |   |   | 2        |   |   |   |   |   |   |   |   |   | 3 |   |  |  |  |  |  |  |  |  |
|-----------------------------------------------------|---|---|---|---|---|---|---|---|---|------|---|---|---|---|---|---|---|---|---|----------|---|---|---|---|---|---|---|---|---|---|---|--|--|--|--|--|--|--|--|
| 0                                                   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0        | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | 1 |  |  |  |  |  |  |  |  |
| Type                                                |   |   |   |   |   |   |   |   |   | Code |   |   |   |   |   |   |   |   |   | Checksum |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |  |  |  |  |
| unused                                              |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |          |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |  |  |  |  |
| Internet Header + 64 bits of Original Data Datagram |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |          |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |  |  |  |  |

# Demo2-ICMPv4 Time Exceeded(1)



Build the network as the topology shown on the top.

Do the configuration to make PC1 could send/receive the packet to/from PC2.

Do the capture on the Ethernet 0/0/0 of PC1

Initiate “**tracert**” testing on PC1 to track the route information on the network path from PC1 to PC2.

Answer the following question:

Q1. What’s the type of the ICMP message would be received by PC1?

Q2. How does PC1 get the route information on the network path from PC1 to PC2?

Q3. In this test, Which interfaces on the network path can a PC obtain relevant information from ?

```
PC> tracert 192.168.3.101
```

```
1 ? xxms xxms xxms
```

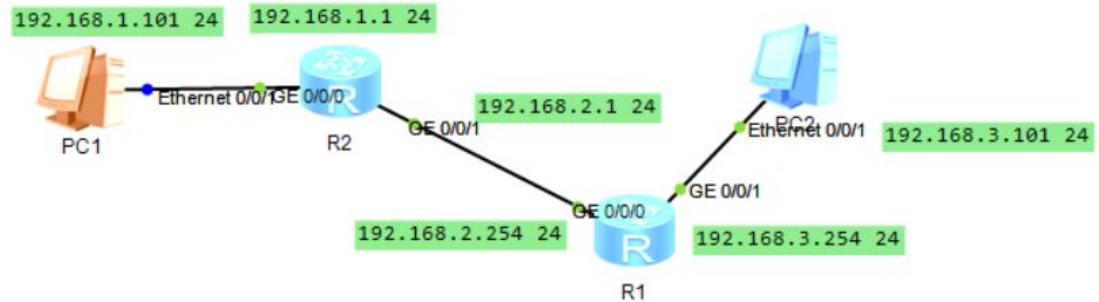
```
2 ? xxms xxms xxms
```

```
3 ? xxms xxms xxms
```

# Demo2- ICMPv4 Time Exceeded(2)

Do the capture on the Ethernet 0/0/0 of PC1

Initiate “**tracert**” testing on PC1 to track the route information on the network path from PC1 to PC2.

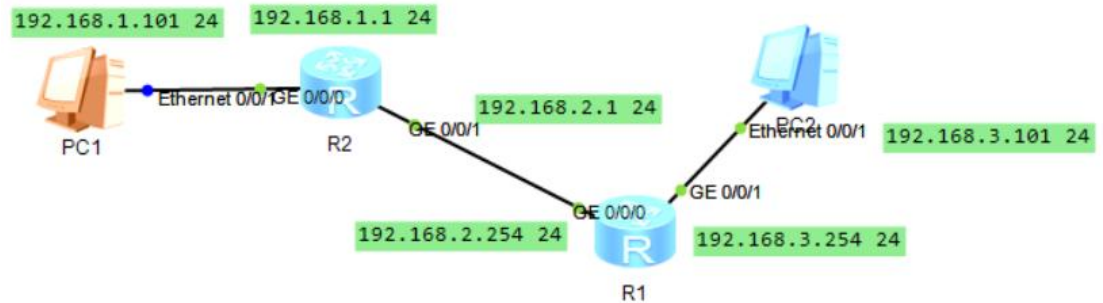


```
PC1
Basic Config | Command | MCPacket | UdpPacket | Console
PC>tracert 192.168.3.101
tracert to 192.168.3.101, 8 hops max
(ICMP), press Ctrl+C to stop
 1 192.168.1.1 32 ms 15 ms 16 ms
 2 192.168.2.254 47 ms 47 ms 46 ms
 3 192.168.3.101 63 ms 62 ms 79 ms
PC>
```

|               |      |               |                                                                     |
|---------------|------|---------------|---------------------------------------------------------------------|
| 192.168.1.101 | ICMP | 192.168.3.101 | 106 Echo (ping) request id=0x5562, seq=1/256, ttl=1 (no response)   |
| 192.168.1.1   | ICMP | 192.168.1.101 | 70 Time-to-live exceeded (Time to live exceeded in transit)         |
| 192.168.1.101 | ICMP | 192.168.3.101 | 106 Echo (ping) request id=0x5562, seq=2/512, ttl=1 (no response)   |
| 192.168.1.1   | ICMP | 192.168.1.101 | 70 Time-to-live exceeded (Time to live exceeded in transit)         |
| 192.168.1.101 | ICMP | 192.168.3.101 | 106 Echo (ping) request id=0x5562, seq=3/768, ttl=1 (no response)   |
| 192.168.1.1   | ICMP | 192.168.1.101 | 70 Time-to-live exceeded (Time to live exceeded in transit)         |
| 192.168.1.101 | ICMP | 192.168.3.101 | 106 Echo (ping) request id=0x5562, seq=1/256, ttl=2 (no response)   |
| 192.168.2.254 | ICMP | 192.168.1.101 | 70 Time-to-live exceeded (Time to live exceeded in transit)         |
| 192.168.1.101 | ICMP | 192.168.3.101 | 106 Echo (ping) request id=0x5562, seq=2/512, ttl=2 (no response)   |
| 192.168.2.254 | ICMP | 192.168.1.101 | 70 Time-to-live exceeded (Time to live exceeded in transit)         |
| 192.168.1.101 | ICMP | 192.168.3.101 | 106 Echo (ping) request id=0x5662, seq=3/768, ttl=2 (no response)   |
| 192.168.2.254 | ICMP | 192.168.1.101 | 70 Time-to-live exceeded (Time to live exceeded in transit)         |
| 192.168.1.101 | ICMP | 192.168.3.101 | 106 Echo (ping) request id=0x5662, seq=1/256, ttl=3 (reply in 66)   |
| 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) reply id=0x5662, seq=1/256, ttl=126 (request in 66) |
| 192.168.1.101 | ICMP | 192.168.3.101 | 106 Echo (ping) request id=0x5662, seq=2/512, ttl=3 (reply in 68)   |
| 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) reply id=0x5662, seq=2/512, ttl=126 (request in 68) |
| 192.168.1.101 | ICMP | 192.168.3.101 | 106 Echo (ping) request id=0x5662, seq=3/768, ttl=3 (reply in 70)   |
| 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) reply id=0x5662, seq=3/768, ttl=126 (request in 70) |

# Demo2- ICMPv4 Time Exceeded(3)

```
PC1
Basic Config Command MCPacket UdpPacket Console
PC>tracert 192.168.3.101
tracert to 192.168.3.101, 8 hops max
(ICMP), press Ctrl+C to stop
 1 192.168.1.1 32 ms 15 ms 16 ms
 2 192.168.2.254 47 ms 47 ms 46 ms
 3 192.168.3.101 63 ms 62 ms 79 ms
PC>
```



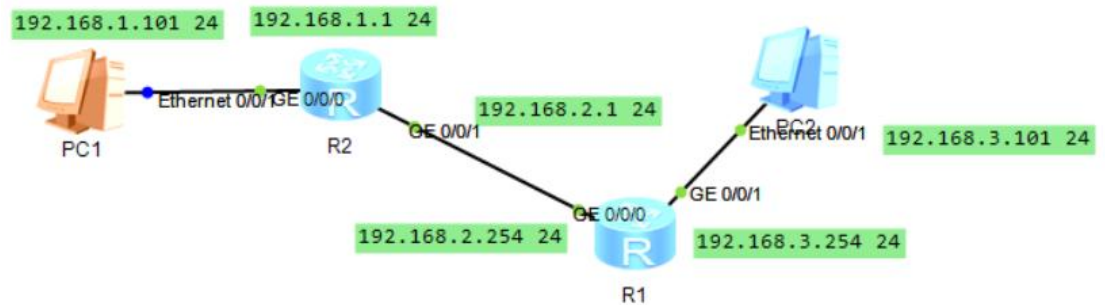
```
PC>ipconfig
Link local IPv6 address.....: fe80::5689:98ff:fec8:5b
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 192.168.1.101
Subnet mask.....: 255.255.255.0
Gateway.....: 192.168.1.1
Physical address.....: 54-89-98-C8-5B-6D
DNS server.....:
```

The destination is not in the same subnet as the source, the packet reaches to the 1st “next hop” : gateway.

```
54 729.969000 192.168.1.1 ICMP 192.168.1.101 70 Time-to-live exceeded
> Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.101
 > Internet Control Message Protocol
 Type: 11 (Time-to-live exceeded)
 Code: 0 (Time to live exceeded in transit)
 Checksum: 0xda04 [correct]
 [Checksum Status: Good]
 Unused: 00000000
 > Internet Protocol Version 4, Src: 192.168.1.101, Dst: 192.168.3.101
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 92
 Identification: 0x6255 (25173)
 > 010. = Flags: 0x2, Don't fragment
 ... 0 0000 0000 0000 Fragment Offset: 0
 > Time to Live: 1
 Protocol: ICMP (1)
 Header Checksum: 0x9131 [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 192.168.1.101
 Destination Address: 192.168.3.101
 > Internet Control Message Protocol
```

# Demo2- ICMPv4 Time Exceeded(4)

```
PC1
Basic Config Command MCPacket UdpPacket Cons
PC>tracert 192.168.3.101
tracert to 192.168.3.101, 8 hops max
(ICMP), press Ctrl+C to stop
 1 192.168.1.1 32 ms 15 ms 16 ms
 2 192.168.2.254 47 ms 47 ms 46 ms
 3 192.168.3.101 63 ms 62 ms 79 ms
PC>
```



R2

```
<Huawei>dis ip routing-table
Route Flags: R - relay, D - download to fib

Routing Tables: Public
Destinations : 7 Routes : 7
```

| Destination/Mask | Proto  | Pre | Cost | Flags | NextHop       | Interface       |
|------------------|--------|-----|------|-------|---------------|-----------------|
| 127.0.0.0/8      | Direct | 0   | 0    | D     | 127.0.0.1     | InLoopBack0     |
| 127.0.0.1/32     | Direct | 0   | 0    | D     | 127.0.0.1     | InLoopBack0     |
| 192.168.1.0/24   | Direct | 0   | 0    | D     | 192.168.1.1   | GigabitEthernet |
| 0/0/0            |        |     |      |       |               |                 |
| 192.168.1.1/32   | Direct | 0   | 0    | D     | 127.0.0.1     | GigabitEthernet |
| 0/0/0            |        |     |      |       |               |                 |
| 192.168.2.0/24   | Direct | 0   | 0    | D     | 192.168.2.1   | GigabitEthernet |
| 0/0/1            |        |     |      |       |               |                 |
| 192.168.2.1/32   | Direct | 0   | 0    | D     | 127.0.0.1     | GigabitEthernet |
| 0/0/1            |        |     |      |       |               |                 |
| 192.168.3.0/24   | Static | 60  | 0    | RD    | 192.168.2.254 | GigabitEthernet |
| 0/0/1            |        |     |      |       |               |                 |

The packet reaches to the 2nd “next hop”:  
192.168.2.254

```
60 730.047000 192.168.2.254 ICMP 192.168.1.101 70 Time-to-live exceeded (Time to live exceeded in transit)
```

```
Internet Protocol Version 4, Src: 192.168.2.254, Dst: 192.168.1.101
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 > Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
 Total Length: 56
 Identification: 0x0015 (21)
 > 000. = Flags: 0x0
 ...0 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 254
 Protocol: ICMP (1)
 Header Checksum: 0x363c [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 192.168.2.254
 Destination Address: 192.168.1.101

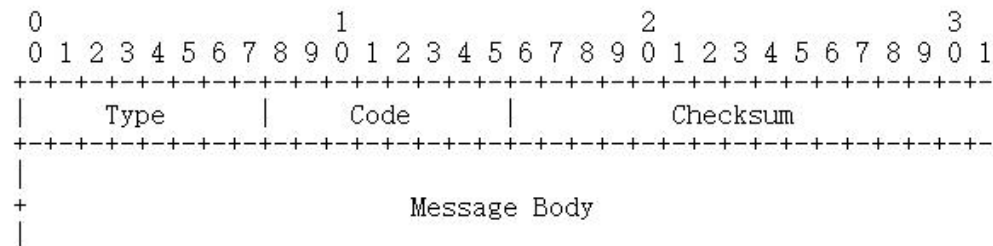
Internet Control Message Protocol
 Type: 11 (Time-to-live exceeded)
 Code: 0 (Time to live exceeded in transit)
 Checksum: 0xda04 [correct]
 [Checksum Status: Good]
 Unused: 00000000

Internet Protocol Version 4, Src: 192.168.1.101, Dst: 192.168.3.101
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 92
 Identification: 0x6258 (25176)
 > 010. = Flags: 0x2, Don't fragment
 ...0 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 1
 Protocol: ICMP (1)
 Header Checksum: 0x912e [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 192.168.1.101
```

# ICMPv6

- **ICMPv6** is used by IPv6 nodes to report errors encountered in processing packets, and to perform other internet-layer functions, such as diagnostics (ICMPv6 "ping").
- ICMPv6 messages are grouped into two classes: error messages and informational messages.
  - Error messages are identified as such by a zero in the high-order bit of their message Type field values.
    - **Error messages have message types from 0 to 127;**
    - **Informational messages have message types from 128 to 255.**

The ICMPv6 messages have the following general format:



<https://www.rfc-editor.org/rfc/rfc4443.html>

# ICMPv6 - examples

```

49 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) request id=0x0001,
<
> Frame 49: 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface \Device\NPF_{7...
Ethernet II, Src: IntelCor_d2:20:30 (04:ed:33:d2:20:30), Dst: JuniperN...
Internet Protocol Version 6, Src: 2001:da8:201d:1109::fcd7, Dst: 2001:c...
Internet Control Message Protocol v6
 Type: Echo (ping) request (128)
 Code: 0
 Checksum: 0xd5bc [correct]
 [Checksum Status: Good]
 Identifier: 0x0001
 Sequence: 13
 [No response seen]
 Data (64 bytes)

```

```

50 ... 2001:da8:20... ICMPv6 2001:da8:2... Time Exceeded (hop limit exceeded in transit)
51 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) request id=0x0001, hop limit=1, (no response)
<
> Frame 50: 174 bytes on wire (1392 bits), 174 bytes captured (1392 bits) on interface \Device\NPF_{7...
Ethernet II, Src: JuniperN_d0:93:c1 (3c:8c:93:d0:93:c1), Dst: IntelCor_d2:20:30 (04:ed:33:d2:20:30)
Internet Protocol Version 6, Src: 2001:da8:201d:1107::1, Dst: 2001:da8:201d:1109::fcd7
Internet Control Message Protocol v6
 Type: Time Exceeded (3)
 Code: 0 (hop limit exceeded in transit)
 Checksum: 0xa810 [correct]
 [Checksum Status: Good]
 Reserved: 00000000
 > Internet Protocol Version 6, Src: 2001:da8:201d:1109::fcd7, Dst: 2001:da8:201d::42:3
 > Internet Control Message Protocol v6

```

```

66 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) reply id=
<
> Frame 66: 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface \Device\NPF_{7...
Ethernet II, Src: JuniperN_d0:93:c1 (3c:8c:93:d0:93:c1), Dst: IntelCor_d2:20:30 (04:ed:33:d2:20:30)
Internet Protocol Version 6, Src: 2001:da8:201d::42:3, Dst: 2001:da8:201d:1109::fcd7
Internet Control Message Protocol v6
 Type: Echo (ping) reply (129)
 Code: 0
 Checksum: 0xd4b6 [correct]
 [Checksum Status: Good]
 Identifier: 0x0001
 Sequence: 19
 [Response To: 65]
 [Response Time: 6.266 ms]
 Data (64 bytes)

```

```

31 10.442477 fe01::101 ICMPv6 fe00::1 Packet Too Big
<
> Frame 31: 1294 bytes on wire (10352 bits), 1294 bytes captured (10352 bits) on interface \Device\NPF_{7...
Ethernet II, Src: HuaweiTe_fc:2a:3d (00:e0:fc:fc:2a:3d), Dst: 02:00:4c:4f:4f:50 (02:00:4c:4f:4f:50)
Internet Protocol Version 6, Src: fe01::101, Dst: fe00::1
Internet Control Message Protocol v6
 Type: Packet Too Big (2)
 Code: 0
 Checksum: 0xf4ce [correct]
 [Checksum Status: Good]
 MTU: 1280
 > Internet Protocol Version 6, Src: fe00::1, Dst: fe02::4
 > Internet Control Message Protocol v6

```

| Error/Info | DETAIL         | TYPE | CODE |
|------------|----------------|------|------|
| Info       | Echo request   | 128  | 0    |
| Info       | Echo reply     | 129  | 0    |
| Error      | Time Exceeded  | 3    | 0    |
| Error      | Packet Too Big | 2    | 0    |
| ...        | ...            | ...  | ...  |

# ICMPv6 - examples(1) Echo request

using 'ping -6 localhost' to invoke an ICMPv6 transaction.

icmpv6.type==128

| No. | Time     | Source | Protocol | Destination | Info                                                             |
|-----|----------|--------|----------|-------------|------------------------------------------------------------------|
| 2   | 3.761658 | ::1    | ICMPv6   | ::1         | Echo (ping) request id=0x0001, seq=5, hop limit=128 (reply in 3) |

> Frame 2: 84 bytes on wire (672 bits), 84 bytes captured (672 bits) on interface \Device\NPF\_Loopback, id 0

> Null/Loopback

> Internet Protocol Version 6, Src: ::1, Dst: ::1

✓ Internet Control Message Protocol v6

Type: Echo (ping) request (128)

Code: 0

Checksum: 0xd4f1 [correct]

[Checksum Status: Good]

Identifier: 0x0001

Sequence: 5

[\[Response In: 3\]](#)

✓ Data (32 bytes)

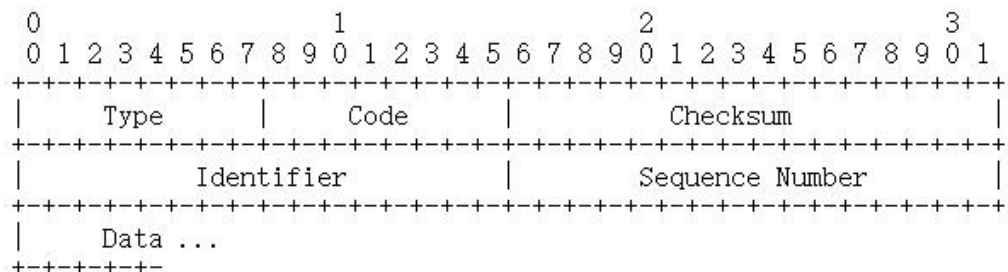
Data: 6162636465666768696a6b6c6d6e6f7071727374757677616263646566676869

[Length: 32]

ICMPv6. type : 128

ICMPv6. code : 0

ICMPv6 Echo request



# ICMPv6- examples(1) Echo reply

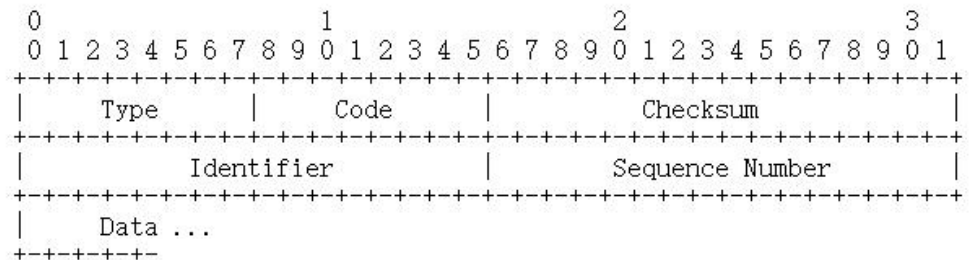
using 'ping -6 localhost' to invoke an ICMPv6 transaction.

| icmpv6.type==129                                                                                             |          |        |          |             |                                                                  |
|--------------------------------------------------------------------------------------------------------------|----------|--------|----------|-------------|------------------------------------------------------------------|
| No.                                                                                                          | Time     | Source | Protocol | Destination | Info                                                             |
| 3                                                                                                            | 3.761717 | ::1    | ICMPv6   | ::1         | Echo (ping) reply id=0x0001, seq=5, hop limit=128 (request in 2) |
| <                                                                                                            |          |        |          |             |                                                                  |
| > Frame 3: 84 bytes on wire (672 bits), 84 bytes captured (672 bits) on interface \Device\NPF_Loopback, id 0 |          |        |          |             |                                                                  |
| > Null/Loopback                                                                                              |          |        |          |             |                                                                  |
| > Internet Protocol Version 6, Src: ::1, Dst: ::1                                                            |          |        |          |             |                                                                  |
| v Internet Control Message Protocol v6                                                                       |          |        |          |             |                                                                  |
| Type: Echo (ping) reply (129)                                                                                |          |        |          |             |                                                                  |
| Code: 0                                                                                                      |          |        |          |             |                                                                  |
| Checksum: 0xd3f1 [correct]                                                                                   |          |        |          |             |                                                                  |
| [Checksum Status: Good]                                                                                      |          |        |          |             |                                                                  |
| Identifier: 0x0001                                                                                           |          |        |          |             |                                                                  |
| Sequence: 5                                                                                                  |          |        |          |             |                                                                  |
| <a href="#">[Response To: 2]</a>                                                                             |          |        |          |             |                                                                  |
| [Response Time: 0.059 ms]                                                                                    |          |        |          |             |                                                                  |
| v Data (32 bytes)                                                                                            |          |        |          |             |                                                                  |
| Data: 6162636465666768696a6b6c6d6e6f7071727374757677616263646566676869                                       |          |        |          |             |                                                                  |
| [Length: 32]                                                                                                 |          |        |          |             |                                                                  |

ICMPv6. type : 129

ICMPv6. code : 0

ICMPv6 Echo reply



# ICMPv6- examples(2) Time Exceeded

using 'tracert -6 www.sustech.edu.cn' to invoke an ICMPv6 transaction.

```
49 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) request id=0x0001, seq=13, hop limit=1 (no response found!)
50 ... 2001:da8:20... ICMPv6 2001:da8:2... Time Exceeded (hop limit exceeded in transit)
51 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) request id=0x0001, seq=14, hop limit=1 (no response found!)
52 ... 2001:da8:20... ICMPv6 2001:da8:2... Time Exceeded (hop limit exceeded in transit)
53 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) request id=0x0001, seq=15, hop limit=1 (no response found!)
54 ... 2001:da8:20... ICMPv6 2001:da8:2... Time Exceeded (hop limit exceeded in transit)
58 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) request id=0x0001, seq=16, hop limit=2 (no response found!)
59 ... 2001:da8:20... ICMPv6 2001:da8:2... Time Exceeded (hop limit exceeded in transit)
60 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) request id=0x0001, seq=17, hop limit=2 (no response found!)
61 ... 2001:da8:20... ICMPv6 2001:da8:2... Time Exceeded (hop limit exceeded in transit)
62 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) request id=0x0001, seq=18, hop limit=2 (no response found!)
63 ... 2001:da8:20... ICMPv6 2001:da8:2... Time Exceeded (hop limit exceeded in transit)
65 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) request id=0x0001, seq=19, hop limit=3 (reply in 66)
66 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) reply id=0x0001, seq=19, hop limit=62 (request in 65)
67 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) request id=0x0001, seq=20, hop limit=3 (reply in 68)
68 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) reply id=0x0001, seq=20, hop limit=62 (request in 67)
69 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) request id=0x0001, seq=21, hop limit=3 (reply in 70)
70 ... 2001:da8:20... ICMPv6 2001:da8:2... Echo (ping) reply id=0x0001, seq=21, hop limit=62 (request in 69)
```

# Practise 11.1

Use the “ping” command to initiate an echo session for ICMPv4, and use Wireshark to capture the following packets. What is the parameter value specified by -l in the following command: “ping www.example.cn -l \_?\_”

Wireshark packet capture showing two ICMP echo request packets. The first packet (No. 1) is fragmented, and the second packet (No. 2) is not fragmented. Red boxes highlight the 'Don't fragment' flag in both packets. Red arrows point to the 'Fragment offset' field in both packets.

| No. | Time     | Source        | Destination | Protocol | Length | Info                                                            |
|-----|----------|---------------|-------------|----------|--------|-----------------------------------------------------------------|
| 1   | 0.000000 | 192.168.2.104 | 47.75.42.25 | IPv4     | 1500   | Fragmented IP protocol (proto=ICMP 1, off=0, ID=e6be)           |
| 2   | 0.000000 | 192.168.2.104 | 47.75.42.25 | ICMP     | 62     | Echo (ping) request id=0x0001, seq=29/7424, ttl=64 (reply in 4) |

Frame 1: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \Device\NPF\_{...} 1514 bytes captured on interface \Device\NPF\_{...}

Ethernet II, Src: IntelCor\_5..., Dst: Skyworth\_de:ad:05 (00:1a:9a:de:ad:05)

Internet Protocol Version 4, Src: 192.168.2.104, Dst: 47.75.42.25

0100 .... = Version: 4  
... 0101 = Header Length: 20 bytes (5)  
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)  
Total Length: 1500  
Identification: 0xe6be (59070)  
Flags: 0x2000, More fragments  
0... .. = Reserved bit: Not set  
.0.. .. = Don't fragment: Not set  
..1. .... = More fragments: Set  
Fragment offset: 0  
Time to live: 64  
Protocol: ICMP (1)  
Header checksum: 0x51ee [validation disabled]  
[Header checksum status: Unverified]  
Source: 192.168.2.104  
Destination: 47.75.42.25  
[Reassembled IPv4 in frame: 2]  
Data (1480 bytes)

Frame 2: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface \Device\NPF\_{...} 62 bytes captured on interface \Device\NPF\_{...}

Ethernet II, Src: IntelCor\_5..., Dst: Skyworth\_de:ad:05 (00:1a:9a:de:ad:05)

Internet Protocol Version 4, Src: 192.168.2.104, Dst: 47.75.42.25

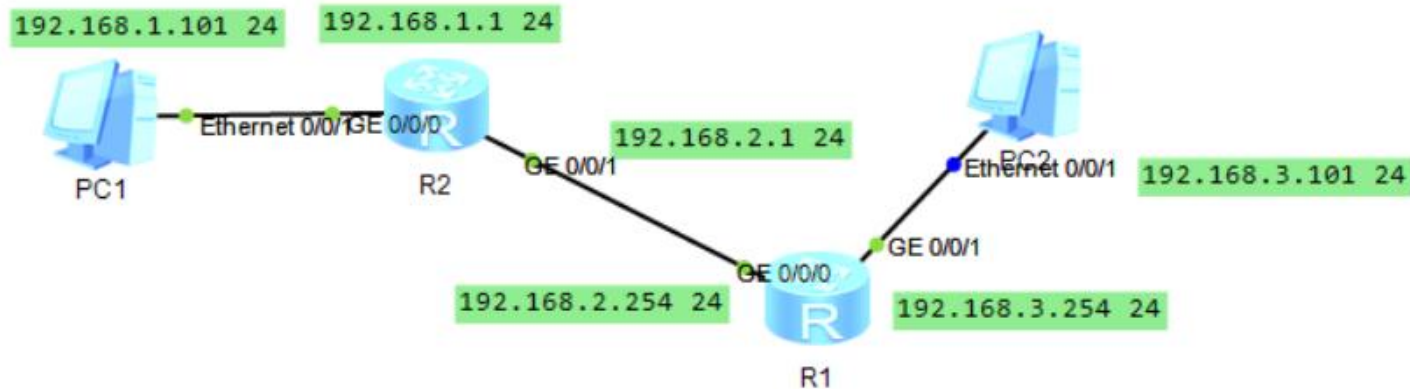
0100 .... = Version: 4  
... 0101 = Header Length: 20 bytes (5)  
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)  
Total Length: 48  
Identification: 0xe6be (59070)  
Flags: 0x00b9  
0... .. = Reserved bit: Not set  
.0.. .. = Don't fragment: Not set  
..0. .... = More fragments: Not set  
Fragment offset: 1480  
Time to live: 64  
Protocol: ICMP (1)  
Header checksum: 0x76e1 [validation disabled]  
[Header checksum status: Unverified]  
Source: 192.168.2.104  
Destination: 47.75.42.25  
[2 IPv4 Fragments (1508 bytes): #1(1480), #2(28)]  
Internet Control Message Protocol

# Practise 11.2

1. Complete the following table based on the differences between ICMPv4 Echo Request and ICMPv6 Echo Request, and compare and explain the ICMPv4 and ICMPv6 packets captured by Wireshark.
2. Based on the captured ICMPv4 or ICMPv6 packets , calculate (programmed or manually calculate) the checksum field in the message, determine whether the calculated checksum value is consistent with the 'checksum' field value in the header of the ICMPv4 or ICMPv6 packets , and demonstrate your calculation process.

|                                          | ICMPv4 | ICMPv6 |
|------------------------------------------|--------|--------|
| Protocol Number                          |        |        |
| Position in IP packet                    |        |        |
| The value of 'type' field in ICMP header |        |        |
| The calculation method of checksum       |        |        |
| The calculation range of checksum        |        |        |
| function                                 |        |        |

# Practise 11.3(1)



Build the network as the topology shown on the top, do the configuration to make PC1 could send/receive packet to/from PC2.

- Use “tracert” to trace the route information from PC2 to PC1, and capture the packets while tracing.
- Please fill in the values of A, B, and C in the table below based on your test, and answer the following questions:

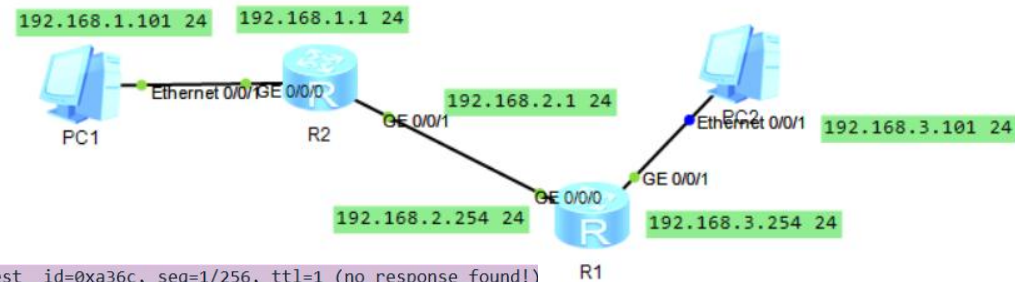
PC> tracert 192.168.1.101

1 A xxms xxms xxms  
2 B xxms xxms xxms  
3 C xxms xxms xxms  
?

Q1. What's the relationship between the three addresses “A”, “B”, and “C” and the following items ?

- 1) the gateway of PC1
- 2) the gateway of PC2
- 3) the next-hop of routing entry on R1 to subnet(192.168.1.0 24)
- 4) the next-hop of routing entry on R2 to subnet(192.168.3.0 24)

# Practise 11.3(2)



|    |           |               |      |               |                                                                          |
|----|-----------|---------------|------|---------------|--------------------------------------------------------------------------|
| 12 | 17.781000 | 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) request id=0xa36c, seq=1/256, ttl=1 (no response found!) |
| 13 | 17.797000 |               | ICMP | 192.168.3.101 | 70 Time-to-live exceeded (Time to live exceeded in transit)              |
| 14 | 17.813000 | 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) request id=0xa36c, seq=2/512, ttl=1 (no response found!) |
| 15 | 17.813000 |               | ICMP | 192.168.3.101 | 70 Time-to-live exceeded (Time to live exceeded in transit)              |
| 16 | 17.828000 | 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) request id=0xa46c, seq=3/768, ttl=1 (no response found!) |
| 17 | 17.844000 |               | ICMP | 192.168.3.101 | 70 Time-to-live exceeded (Time to live exceeded in transit)              |
| 18 | 17.859000 | 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) request id=0xa46c, seq=1/256, ttl=2 (no response found!) |
| 19 | 17.891000 |               | ICMP | 192.168.3.101 | 70 Time-to-live exceeded (Time to live exceeded in transit)              |
| 20 | 17.891000 | 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) request id=0xa46c, seq=2/512, ttl=2 (no response found!) |
| 21 | 17.938000 |               | ICMP | 192.168.3.101 | 70 Time-to-live exceeded (Time to live exceeded in transit)              |
| 22 | 17.938000 | 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) request id=0xa46c, seq=3/768, ttl=2 (no response found!) |
| 23 | 17.984000 |               | ICMP | 192.168.3.101 | 70 Time-to-live exceeded (Time to live exceeded in transit)              |
| 24 | 17.984000 | 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) request id=0xa46c, seq=1/256, ttl=3 (reply in 25)        |
| 25 | 18.047000 |               | ICMP | 192.168.3.101 | 106 Echo (ping) reply id=0xa46c, seq=1/256, ttl=126 (request in 24)      |
| 26 | 18.047000 | 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) request id=0xa46c, seq=2/512, ttl=3 (reply in 27)        |
| 27 | 18.109000 |               | ICMP | 192.168.3.101 | 106 Echo (ping) reply id=0xa46c, seq=2/512, ttl=126 (request in 26)      |
| 28 | 18.125000 | 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) request id=0xa46c, seq=3/768, ttl=3 (reply in 29)        |
| 29 | 18.172000 |               | ICMP | 192.168.3.101 | 106 Echo (ping) reply id=0xa46c, seq=3/768, ttl=126 (request in 28)      |

Q2. What is the value of the “TTL” field in the IP header of the following numbered IP packet: 12, 14, 16, 18, 20, 22, 24, 26, 28?

Tips: here #12 is the first ICMP echo request message sent by PC2 during the “tracert” test.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |               |      |               |                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|------|---------------|--------------------------------------------------|
| 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 17.781000 | 192.168.3.101 | ICMP | 192.168.1.101 | 106 Echo (ping) request id=0xa36c, seq=1/256, tt |
| <p>&gt; Frame 12: 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface -, id 0</p> <p>&gt; Ethernet II, Src: HuaweiE_80:74:07 (54:89:98:80:74:07), Dst: HuaweiE_9b:2e:3c (54:89:98:9b:2e:3c)</p> <p>&gt; Internet Protocol Version 4, Src: 192.168.3.101, Dst: 192.168.1.101</p> <p>0100 .... = Version: 4</p> <p>.... 0101 = Header Length: 20 bytes (5)</p> <p>&gt; Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)</p> <p>Total Length: 92</p> <p>Identification: 0x6ca3 (27811)</p> <p>&gt; 010. .... = Flags: 0x2, Don't fragment</p> <p>...0 0000 0000 0000 = Fragment Offset: 0</p> <p>&gt; Time to Live: 1</p> <p>Protocol: ICMP (1)</p> <p>Header Checksum: 0x86e3 [validation disabled]</p> <p>[Header checksum status: Unverified]</p> <p>Source Address: 192.168.3.101</p> <p>Destination Address: 192.168.1.101</p> <p>&gt; Internet Control Message Protocol</p> |           |               |      |               |                                                  |

what's the value of Time to Live field in the IP header ?